

Module 1: Number Systems — Answer Key

Prequiz

1. -35 is an **integer**. It is not natural, and integers are contained in both the rationals and reals, so $\mathbb{Z}$ is the smallest standard set listed that contains it.

2. The trip is

$$-6 + 10 + (-14) = -10.$$

The final position is $\boxed{-10}$.

3. Cross-products:

$$6 \cdot 21 = 126, \quad 9 \cdot 14 = 126,$$

so the fractions are equivalent. Also,

$$\frac{6}{9} = \frac{2}{3}, \quad \frac{14}{21} = \frac{2}{3}.$$

4. A suitable recursion is

$$H_0 = 800, \quad H_{n+1} = H_n + 120.$$

After four years,

$$H_4 = 800 + 4(120) = \boxed{1280}.$$

- 5.

$$B_0 = 800, \quad B_{n+1} = 1.10B_n.$$

Then

$$B_3 = 800(1.10)^3 = \boxed{1064.8}.$$

The **ratio** is constant: each year's value is 1.10 times the previous year's value.

Matching quiz

1. -8 is an **integer**.

- 2.

$$-3 + 11 + (-6) = \boxed{2}.$$

- 3.

$$\frac{8}{12} = \frac{2}{3}, \quad \frac{18}{27} = \frac{2}{3},$$

so the fractions are equivalent.

- 4.

$$H_0 = 1200, \quad H_{n+1} = H_n + 75,$$

so

$$H_5 = 1200 + 5(75) = \boxed{1575}.$$

- 5.

$$B_0 = 1200, \quad B_{n+1} = 1.05B_n,$$

so

$$B_2 = 1200(1.05)^2 = \boxed{1323}.$$